WORKED SOLUTION

STEP II 1994, Question 8

Pure • Optimising a Quadratic Model

STEP II 1994, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q8

The key idea

Follow the money through one delivery cycle. Stock falls at a steady rate r until it runs out, so $S(t)$ is a simple ramp, and the storage bill is b times the area under that ramp. The profit per cycle is then a quadratic in X. Maximising it over the allowed range $0 < X \leq rT$ gives the condition, and the hypothesis $T \geq p/b$ is exactly what puts the vertex of the parabola inside that range.

Solution

Why $X \leq rT$

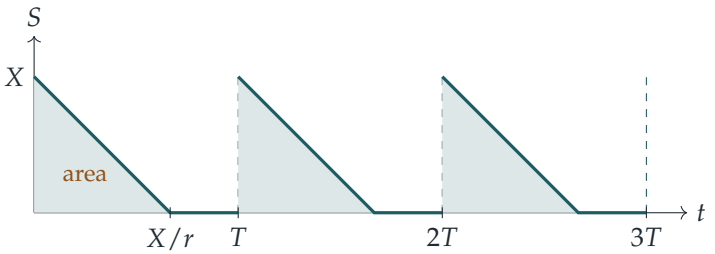
In T hours she can sell at most rT units. If $X > rT$, at least $X - rT$ units are still on the shelf when the next lorry arrives. Each cycle then adds at least $X - rT$ more to this unsold surplus, so after n cycles it is at least $n(X - rT)$. That grows without bound, and so does the storage bill of b per unit per hour. She pays for these units (and for storing them) but they bring in no extra sales, because sales are capped at rT per cycle anyway. So ordering more than rT per delivery only adds cost, and she should take $X \leq rT$.

The stock level

With $X \leq rT$ each cycle starts with exactly X units. In the continuous model she sells at rate r while there's stock, so

$$S(t) = \begin{cases} X - rt, & 0 \leq t \leq X/r, \\ 0, & X/r \leq t \leq T. \end{cases}$$

The graph is a straight line falling from X to zero at $t = X/r$, then flat at zero until the next delivery at $t = T$ (where it jumps back up to X). This gives a sawtooth over several cycles.



Profit per cycle

Over one period of T hours:

- *Sales.* All X units are sold (since $X \le rT$), at $£(p+q)$ each, giving $£(p+q)X$.
- *Delivery.* $£(a+qX)$.
- *Storage.* Storing $S(t)$ units for a short time δt costs $bS(t)\,\delta t$, so the total is

$$b\int_0^T S(t)\,\mathrm{d}t = b\int_0^{X/r}(X-rt)\,\mathrm{d}t = b\cdot\tfrac{1}{2}\cdot X\cdot\frac{X}{r} = \frac{bX^2}{2r},$$

  the area of the shaded triangle times b.
- *Running costs.* $£cT$.

The qX terms cancel, so the profit per cycle is

$$P(X) = pX - \frac{bX^2}{2r} - a - cT.$$

So p is the real profit margin on each unit: q is simply passed on to the customer.

When can the business be profitable?

With T fixed, she controls X in the range $0 < X \le rT$. Completing the square,

$$P(X) = \frac{p^2 r}{2b} - a - cT - \frac{b}{2r}\left(X - \frac{pr}{b}\right)^2.$$

Ask yourself Is the best X actually allowed?

The parabola peaks at $X = pr/b$, and we need $pr/b \le rT$, which is $T \ge p/b$. That's precisely the hypothesis. So she can choose $X = pr/b$, and then

$$P = \frac{p^2 r}{2b} - (a + cT).$$

This is positive exactly when $p^2 r > 2b(a+cT)$, that is, when

$$p^2 > 2\,\frac{(a+cT)b}{r}.$$

So if this inequality holds, the order size $X = pr/b$ makes the business profitable, as required. (In fact pr/b is the best possible choice, so the condition is also necessary when $T \geq p/b$.)

If instead $T < p/b$, the vertex lies beyond rT. Then P is increasing on the whole allowed range and the best she can do is $X = rT$. That's why the hypothesis on T is needed for this particular condition.

Checks

- Take $a = 10$, $b = 1$, $r = 10$, $p = 4$, $c = 1$, $T = 5$. Then $T \geq p/b = 4$, and $p^2 = 16 > 2(15)(1)/10 = 3$. The best order is $X = 40 \leq rT = 50$, with profit $160 - 10 - 80 - 5 = 65 = \frac{16 \cdot 10}{2} - 15$.
- The computer stepped $S(t)$ numerically through a cycle for 30 random sets of parameters and matched $P(X)$ to within 10^{-7}. A grid search over X for 2000 random cases with $T \geq p/b$ confirmed the maximum $\frac{p^2 r}{2b} - a - cT$.
- Units: $p^2 r/b$ is $(£/\text{unit})^2 \times (\text{units/h})/(£/\text{unit/h}) = £$, as are a and cT.

Common slips

- Charging storage on all X units for all T hours (bXT) instead of on the stock actually held, $b \int S \, \mathrm{d}t$.
- Forgetting to check that $X = pr/b$ satisfies $X \leq rT$. That check is the only place the condition $T \geq p/b$ is used.

Beyond STEP

This is the classic *economic order quantity* model of operations research. Balancing a fixed cost per order against a holding cost per unit-time gives a square-root rule. Here, if she can also choose T, the profit *rate* P/T is maximised by trading off a against storage. Constrained optimisation of this kind (here the vertex versus the boundary $X = rT$) grows into linear and nonlinear programming and the Karush–Kuhn–Tucker conditions. For the habit of building and testing such models, Siklos, *Advanced Problems in Mathematics*, is a good companion.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.